ΕΡΓΑΣΤΗΡΙΑΚΗ ΜΕΛΕΤΗ ΙΙΙ
ASSEMBLY 8085

Μ.Ι.Σ. 2021-2022

Να κατασκευαστεί ένα πρόγραμμα για τον επεξεργαστή 8085 το οποίο,

1. να καθαρίζει τις 8bit θέσεις μνήμης με διευθύνσεις 3000_{16} – 3009_{16} και 4000_{16} – 4009_{16},

2. να λάβετε υπόψη την κατασκευή δυο πινάκων, τον X = [*x*0, *x*1, …, *x*9] και τον Y = [*y*0, *y*1, …, *y*9]

3. να γεμίσετε τις 10 θέσεις του πρώτου πίνακα X με τις τιμές 0F,01,1A,02,A1,E1,01,01,04,03

4. να φτιάξετε έναν βρόγχο για τις 10 θέσεις του πίνακα X όπου,

   - εάν ισχύει η έκφραση ($2 < x_{\vartheta\acute{\varepsilon}\sigma\eta_i} <= 6$) τότε να κάνετε την πράξη
     ((4 * $x_{\vartheta\acute{\varepsilon}\sigma\eta_i}$) + $y_{\vartheta\acute{\varepsilon}\sigma\eta_i}$)
     και το αποτέλεσμα της πράξης αυτής, να μπαίνει στην θέση $y_{\vartheta\acute{\varepsilon}\sigma\eta_i}$

   - εάν ΔΕΝ ισχύει η έκφραση ($2 < x_{\vartheta\acute{\varepsilon}\sigma\eta_i} <= 6$) τότε να κάνετε απλά την ανάθεση
     $y_{\vartheta\acute{\varepsilon}\sigma\eta_i} = x_{\vartheta\acute{\varepsilon}\sigma\eta_i}$
     και το αποτέλεσμα να μπαίνει στην θέση $y_{\vartheta\acute{\varepsilon}\sigma\eta_i}$

   - επίσης όταν ικανοποιείται η έκφραση, να αποθηκεύεται στον καταχωρητή B, η επιτυχία με +1

Να κατασκευάσετε αρχικά ένα διάγραμμα ροής για βοήθεια.
Μην λάβετε υπόψη ενέργειες για υπερχείλιση.

Λάβετε υπόψη ότι ο 8085 δεν μπορεί να κάνει πολλαπλασιασμό.
Όμως, ο πολλαπλασιασμός μπορεί να υλοποιηθεί με διαδοχικές προσθέσεις.

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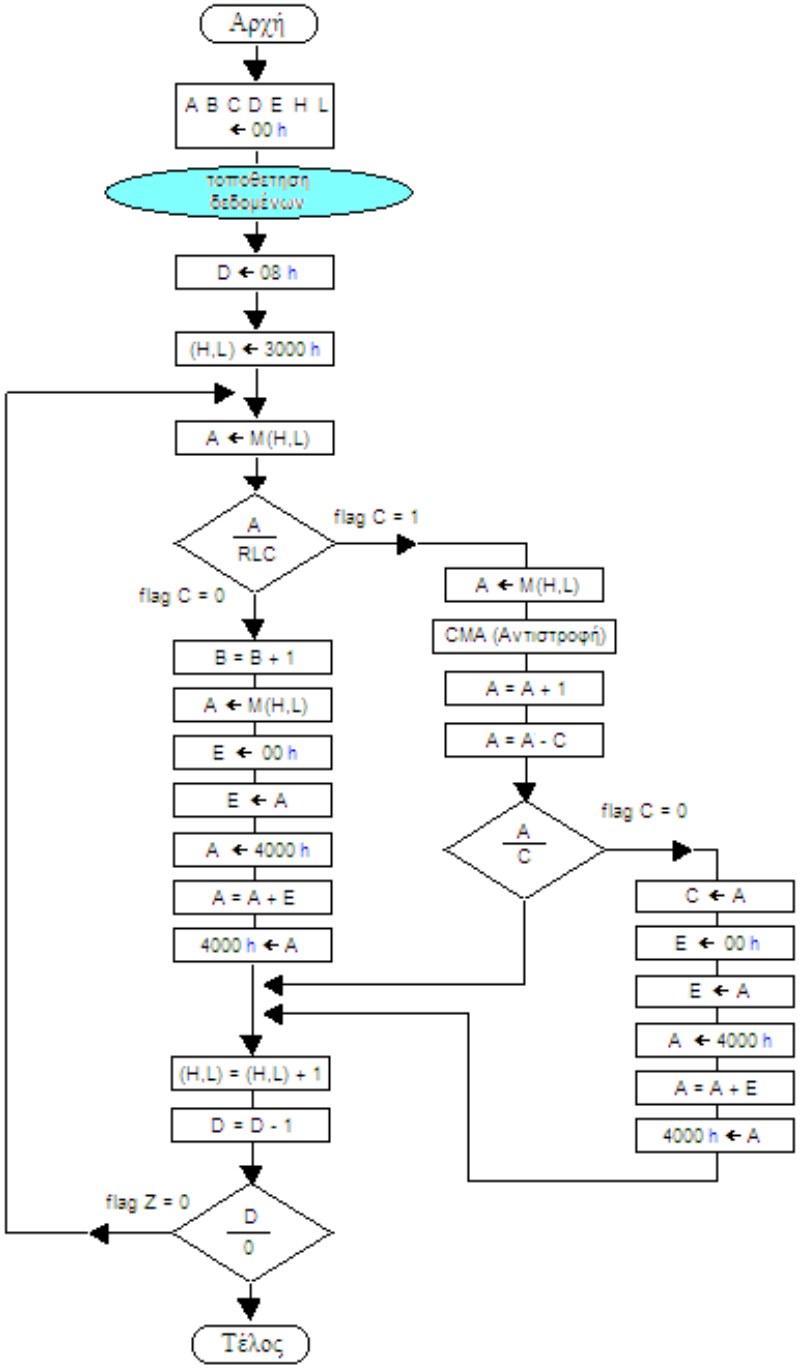
Αρχή
A B C D E H L ← 00 h
τοποθέτηση δεδομένων
D ← 08 h
(H,L) ← 3000 h
A ← M(H,L)
A / RLC
flag C = 1
flag C = 0
A ← M(H,L)
CMA (Αντιστροφή)
A = A + 1
A = A - C
B = B + 1
A ← M(H,L)
E ← 00 h
E ← A
A ← 4000 h
A = A + E
4000 h ← A
A / C
flag C = 0
C ← A
E ← 00 h
E ← A
A ← 4000 h
A = A + E
4000 h ← A
(H,L) = (H,L) + 1
D = D - 1
flag Z = 0
D / 0
Τέλος

```
MVI A,00h
MVI B,00h
MVI C,00h
MVI D,00h
MVI E,00h
MVI H,00h
MVI L,00h

MVI A,EFh
STA 3000h
MVI A,01h
STA 3001h
MVI A,51h
STA 3002h
MVI A,31h
STA 3003h
MVI A,A1h
STA 3004h
MVI A,E1h
STA 3005h
MVI A,01h
STA 3006h
MVI A,01h
STA 3007h
```

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```
    MVI A,00h
    STA 4000h
    MVI A,00h
    STA 4001h

    MVI A,00h
    MVI D,08h
    LXI H,3000h

START-LOGIC:

    MOV  A,M

    RLC
        JNC ADD-POSITIVE
        JC  FIND-MAX-NEGATIVE

    JMP  CHECK-END-LOGIC

ADD-POSITIVE:

    INR  B

    MOV  A,M
```

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```
    MVI  E,00h
    MOV  E,A
    LDA  4000h
    ADD  E
    STA  4000h

    JMP  CHECK-END-LOGIC

POP-MAX-NEGATIVE:

    MOV  C,A

    MVI  E,00h
    MOV  E,A
    LDA  4001h
    ADD  E
    STA  4001h

    JMP  CHECK-END-LOGIC

FIND-MAX-NEGATIVE:

    MOV  A,M

    CMA
```

Μ.Ι.Σ. 2021-2022

```
    INR  A

    CMP  C

        JNC  POP-MAX-NEGATIVE

    JMP  CHECK-END-LOGIC

CHECK-END-LOGIC:

    INX  H

    DCR  D
        JNZ START-LOGIC

    HLT
```

Μ.Ι.Σ. 2021-2022

```
        MVI  A,00h
        MVI  B,00h
        MVI  C,00h
        MVI  D,00h
        MVI  E,00h
        MVI  H,00h
        MVI  L,00h

        MVI  C,0Ah

        LXI  H,3000h
        LXI  D,4000h

START-LOGIC:

        MOV  A,M

        CPI  02h
             JZ  CREATE-KATO-MEROS
             JC  CREATE-KATO-MEROS

        CPI  07h
             JNC CREATE-KATO-MEROS

CREATE-ANO-MEROS:

        RLC
        RLC

        XCHG

        ADD  M
        MOV  M,A

        XCHG

        INR  B

        JMP  CHECK-END-LOGIC

CREATE-KATO-MEROS:
```

CPU Registers

A	01	45	Flags ⇨	S Z - AC - P - CY	0 1 0 0 0 1 0 1
B	02	00	C	Click to toggle flag bits	
D	40	0A	E		
H	30	0A	L	SP FFF0	PC 1036

Simulation

Compiler messages :

Compiling code ...
Compiled successfully
Running program from address 1000 ...
Execution finished

☐ Monitor registers ☑ Monitor memory

$2 < X \leq 6$

System RAM Cols Fill Copy Reset

Address	Data	Address	Data	Address	Data	Address	Data
1000	3E	2000	00	3000	02	4000	02
1001	00	2001	00	3001	05	4001	1A
1002	06	2002	00	3002	08	4002	08
1003	00	2003	00	3003	01	4003	01
1004	0E	2004	00	3004	03	4004	0D
1005	00	2005	00	3005	10	4005	10
1006	16	2006	00	3006	02	4006	02
1007	00	2007	00	3007	01	4007	01
1008	1E	2008	00	3008	07	4008	07
1009	00	2009	00	3009	01	4009	01
100A	26	200A	00	300A	00	400A	00
100B	00	200B	00	300B	00	400B	00
100C	2E	200C	00	300C	00	400C	00
100D	00	200D	00	300D	00	400D	00
100E	0E	200E	00	300E	00	400E	00
100F	0A	200F	00	300F	00	400F	00
1010	21	2010	00	3010	00	4010	00
1011	00	2011	00	3011	00	4011	00
1012	30	2012	00	3012	00	4012	00
1013	11	2013	00	3013	00	4013	00
1014	00	2014	00	3014	00	4014	00
1015	40	2015	00	3015	00	4015	00
1016	7E	2016	00	3016	00	4016	00
1017	FE	2017	00	3017	00	4017	00
1018	02	2018	00	3018	00	4018	00

Execution finished | T states : 797

```
      MVI  A,00h
      MVI  B,00h
      MVI  C,00h
      MVI  D,00h
      MVI  E,00h
      MVI  H,00h
      MVI  L,00h

      MVI  C,0Ah

      LXI  H,3000h
      LXI  D,4000h

START-LOGIC:

      MOV  A,M

      CPI  02h
           JZ  CREATE-KATO-MEROS
           JC  CREATE-KATO-MEROS

      CPI  07h
           JNC CREATE-KATO-MEROS

CREATE-ANO-MEROS:

      RLC
      RLC

      XCHG

      ADD  M
      MOV  M,A

      XCHG

      INR  B

      JMP  CHECK-END-LOGIC

CREATE-KATO-MEROS:
```

Execution finished

CPU Registers

A	08	44	Flags ⇒	S Z - AC - P - CY	0 1 0 0 0 1 0 0
B	03	00	C	Click to toggle flag bits	
D	40	0A	E		
H	30	0A	L	SP FFF0	PC 1036

$2 < X \leq 6$

Simulation

Compiler messages :

Compiling code ...
Compiled successfully
Running program from address 1000 ...
Execution finished

☐ Monitor registers ☑ Monitor memory

System RAM

Cols Fill Copy Reset

Address	Data	Address	Data	Address	Data	Address	Data
1000	3E	2000	00	3000	05	4000	1A
1001	00	2001	00	3001	05	4001	1C
1002	06	2002	00	3002	05	4002	1A
1003	00	2003	00	3003	01	4003	01
1004	0E	2004	00	3004	02	4004	02
1005	00	2005	00	3005	00	4005	00
1006	16	2006	00	3006	08	4006	08
1007	00	2007	00	3007	11	4007	11
1008	1E	2008	00	3008	01	4008	01
1009	00	2009	00	3009	08	4009	08
100A	26	200A	00	300A	00	400A	00
100B	00	200B	00	300B	00	400B	00
100C	2E	200C	00	300C	00	400C	00
100D	00	200D	00	300D	00	400D	00
100E	0E	200E	00	300E	00	400E	00
100F	0A	200F	00	300F	00	400F	00
1010	21	2010	00	3010	00	4010	00
1011	00	2011	00	3011	00	4011	00
1012	30	2012	00	3012	00	4012	00
1013	11	2013	00	3013	00	4013	00
1014	00	2014	00	3014	00	4014	00
1015	40	2015	00	3015	00	4015	00
1016	7E	2016	00	3016	00	4016	00
1017	FE	2017	00	3017	00	4017	00
1018	02	2018	00	3018	00	4018	00

T states : 849

```
        MVI   A,00h
        MVI   B,00h
        MVI   C,00h
        MVI   D,00h
        MVI   E,00h
        MVI   H,00h
        MVI   L,00h

        MVI   A,EFh
        STA   3000h
        MVI   A,01h
        STA   3001h
        MVI   A,51h
        STA   3002h
        MVI   A,31h
        STA   3003h
        MVI   A,A1h
        STA   3004h
        MVI   A,E1h
        STA   3005h
        MVI   A,01h
        STA   3006h
        MVI   A,01h
        STA   3007h

        MVI   A,00h
        STA   4000h
        MVI   A,00h
        STA   4001h

        MVI   A,00h

        MVI   D,08h
        LXI   H,3000h

START-LOGIC:

        MOV   A,M

        RLC
              JNC ADD-POSITIVE
              JC  FIND-MAX-NEGATIVE
```

ΔΕΔΟΜΕΝΑ

CPU Registers

A	85	44	Flags ⇒	S	Z	-	AC	-	P	-	CY
				0	1	0	0	0	1	0	0
B	05	5F	C	Click to toggle flag bits							
D	00	01	E								
H	30	08	L	SP	FFF9	PC	107F				

ΑΠΑΝΤΗΣΗ

ΑΠΑΝΤΗΣΗ

Simulation

Compiler messages :

Compiling code ...
Compiled successfully
Running program from address 1000 ...
Execution finished

☐ Monitor registers ☑ Monitor memory

System RAM Cols Fill Copy Reset

Address	Data	Address	Data	Address	Data	Address	Data
1000	3E	2000	00	3000	EF	4000	85
1001	00	2001	00	3001	01	4001	70
1002	06	2002	00	3002	51	4002	00
1003	00	2003	00	3003	31	4003	00
1004	0E	2004	00	3004	A1	4004	00
1005	00	2005	00	3005	E1	4005	00
1006	16	2006	00	3006	01	4006	00
1007	00	2007	00	3007	01	4007	00
1008	1E	2008	00	3008	00	4008	00
1009	00	2009	00	3009	00	4009	00
100A	26	200A	00	300A	00	400A	00
100B	00	200B	00	300B	00	400B	00
100C	2E	200C	00	300C	00	400C	00
100D	00	200D	00	300D	00	400D	00
100E	3E	200E	00	300E	00	400E	00
100F	EF	200F	00	300F	00	400F	00
1010	32	2010	00	3010	00	4010	00
1011	00	2011	00	3011	00	4011	00
1012	30	2012	00	3012	00	4012	00
1013	3E	2013	00	3013	00	4013	00
1014	01	2014	00	3014	00	4014	00
1015	32	2015	00	3015	00	4015	00
1016	01	2016	00	3016	00	4016	00
1017	30	2017	00	3017	00	4017	00
1018	3E	2018	00	3018	00	4018	00

ΑΠΑΝΤΗΣΗ

ΑΠΑΝΤΗΣΗ

Execution finished T states : 1138